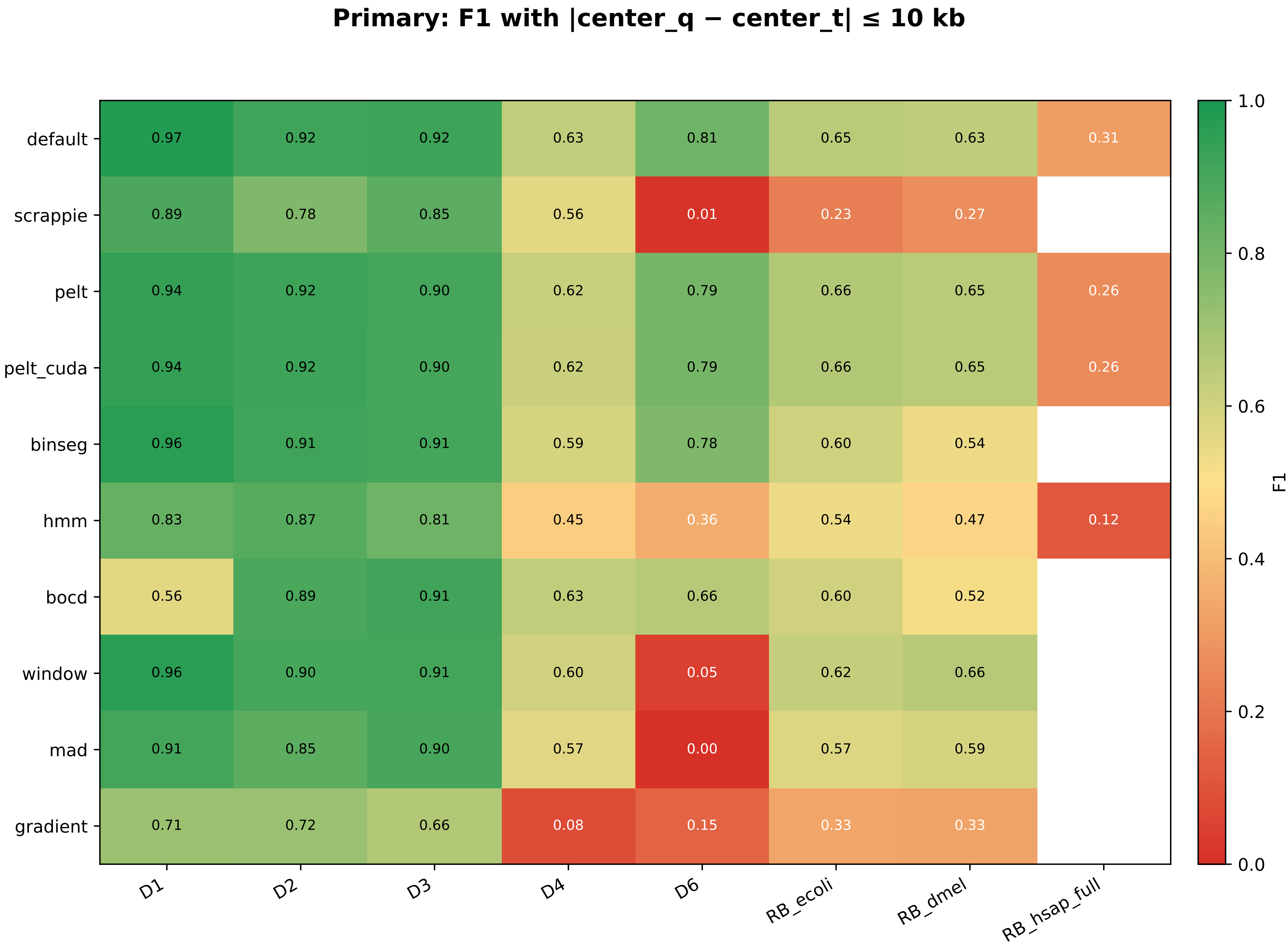
Datasets (basic)

Dataset	Organism	Chemistry	Reads	Ref	FAST5
D1	SARS-CoV-2	R9.4	1,383	31 KB	11.00 GB
D2	E. coli	R9.4	89	5 MB	27.00 GB
D3	S. cerevisiae	R9.4	500	12 MB	0.40 GB
D4	C. reinhardtii	R9.4	500	112 MB	2.00 GB
D6	E. coli	R10.4	294	5 MB	98.00 GB
RB_ecoli	E. coli	R10.4.1	12,000	5 MB	0.93 GB
RB_dmel	D. melanogaster	R10.4.1	12,000	143 MB	0.73 GB
RB_hsap_full	H. sapiens (GRCh38)	R10.4.1	12,000	3.0 GB	7.10 GB

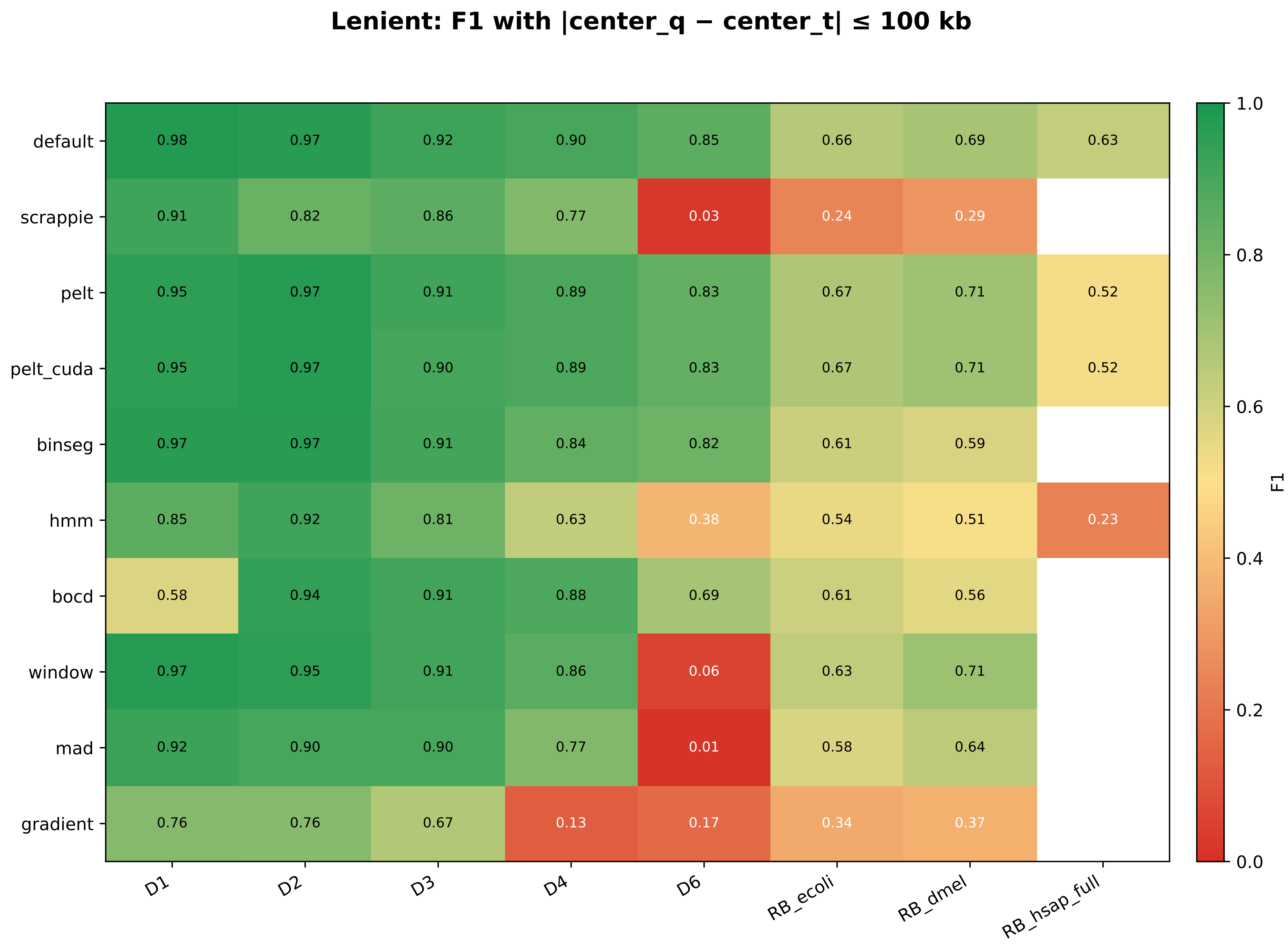
Datasets — extended characterization

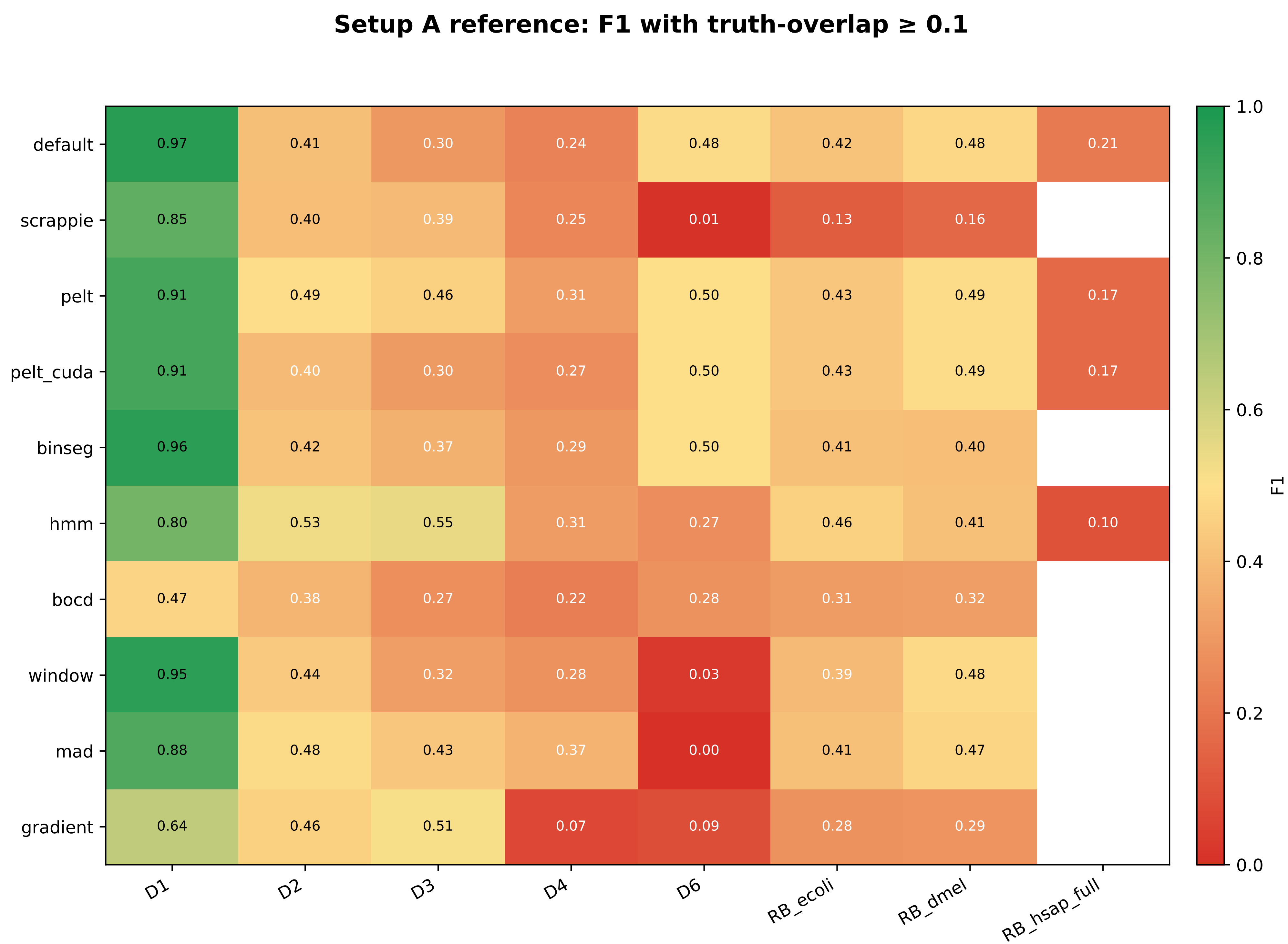
DS	Organism	Chem	Reads	Ref bp	GC%	contigs	N50 (Mb)	N%	masked%
D1	SARS-CoV-2	R9.4	1,383	0.03 Mb	38.0	1	0.03	0.00	0.0
D2	E. coli	R9.4	89	5.23 Mb	50.5	1	5.23	0.00	0.0
D3	S. cerevisiae	R9.4	500	12.16 Mb	38.1	17	0.92	0.00	0.0
D4	C. reinhardtii	R9.4	500	111.10 Mb	64.1	53	7.78	3.65	32.4
D6	E. coli	R10.4	294	5.23 Mb	50.5	1	5.23	0.00	0.0
RB_ecoli	E. coli	R10.4.1	12,000	5.23 Mb	50.5	1	5.23	0.00	0.0
RB_dmel	D. melanogaster	R10.4.1	12,000	143.73 Mb	42.0	1,870	25.29	0.80	20.9
RB_hsap_full	H. sapiens (GRCh3	R10.4.1	12,000	3.21 Gb	41.0	455	145.14	4.98	49.5

Ref bp = total reference length; N50 = contig N50; N% = ambiguous N bases; masked% = soft-masked low-complexity / repeat content (lowercase in FASTA).



Primary: F1 with |center_q − center_t| ≤ 10 kb
default
scrappie
pelt
pelt_cuda
binseg
hmm
bocd
window
mad
gradient
D1
D2
D3
D4
D6
RB_ecoli
RB_dmel
RB_hsap_full
F1
0.0
0.2
0.4
0.6
0.8
1.0
0.97 0.92 0.92 0.63 0.81 0.65 0.63 0.31
0.89 0.78 0.85 0.56 0.01 0.23 0.27
0.94 0.92 0.90 0.62 0.79 0.66 0.65 0.26
0.94 0.92 0.90 0.62 0.79 0.66 0.65 0.26
0.96 0.91 0.91 0.59 0.78 0.60 0.54
0.83 0.87 0.81 0.45 0.36 0.54 0.47 0.12
0.56 0.89 0.91 0.63 0.66 0.60 0.52
0.96 0.90 0.91 0.60 0.05 0.62 0.66
0.91 0.85 0.90 0.57 0.00 0.57 0.59
0.71 0.72 0.66 0.08 0.15 0.33 0.33


Lenient: F1 with |center_q − center_t| ≤ 100 kb
default
scrappie
pelt
pelt_cuda
binseg
hmm
bocd
window
mad
gradient
D1
D2
D3
D4
D6
RB_ecoli
RB_dmel
RB_hsap_full
F1
0.98 0.97 0.92 0.90 0.85 0.66 0.69 0.63
0.91 0.82 0.86 0.77 0.03 0.24 0.29
0.95 0.97 0.91 0.89 0.83 0.67 0.71 0.52
0.95 0.97 0.90 0.89 0.83 0.67 0.71 0.52
0.97 0.97 0.91 0.84 0.82 0.61 0.59
0.85 0.92 0.81 0.63 0.38 0.54 0.51 0.23
0.58 0.94 0.91 0.88 0.69 0.61 0.56
0.97 0.95 0.91 0.86 0.06 0.63 0.71
0.92 0.90 0.90 0.77 0.01 0.58 0.64
0.76 0.76 0.67 0.13 0.17 0.34 0.37
1.0
0.8
0.6
0.4
0.2
0.0


Setup A reference: F1 with truth-overlap ≥ 0.1
default
scrappie
pelt
pelt_cuda
binseg
hmm
bocd
window
mad
gradient
D1
D2
D3
D4
D6
RB_ecoli
RB_dmel
RB_hsap_full
0.97 0.41 0.30 0.24 0.48 0.42 0.48 0.21
0.85 0.40 0.39 0.25 0.01 0.13 0.16
0.91 0.49 0.46 0.31 0.50 0.43 0.49 0.17
0.91 0.40 0.30 0.27 0.50 0.43 0.49 0.17
0.96 0.42 0.37 0.29 0.50 0.41 0.40
0.80 0.53 0.55 0.31 0.27 0.46 0.41 0.10
0.47 0.38 0.27 0.22 0.28 0.31 0.32
0.95 0.44 0.32 0.28 0.03 0.39 0.48
0.88 0.48 0.43 0.37 0.00 0.41 0.47
0.64 0.46 0.51 0.07 0.09 0.28 0.29
F1
0.0
0.2
0.4
0.6
0.8
1.0

Per-dataset F1 across 4 metrics — best per column highlighted

D1 (truth=1,381,999 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.975	0.979	0.968	0.920
scrappie	0.888	0.914	0.847	0.772
pelt	0.938	0.953	0.906	0.671
pelt_cuda	0.938	0.953	0.906	0.671
binseg	0.964	0.968	0.958	0.901
hmm	0.831	0.849	0.802	0.753
bocd	0.560	0.578	0.471	0.278
window	0.964	0.971	0.953	0.907
mad	0.907	0.924	0.881	0.829
gradient	0.713	0.763	0.636	0.571

D2 (truth=350,620 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.915	0.968	0.410	0.071
scrappie	0.777	0.821	0.404	0.066
pelt	0.919	0.972	0.492	0.084
pelt_cuda	0.919	0.972	0.397	0.066
binseg	0.915	0.968	0.417	0.070
hmm	0.866	0.917	0.530	0.096
bocd	0.894	0.945	0.378	0.034
window	0.900	0.953	0.437	0.078
mad	0.851	0.901	0.485	0.086
gradient	0.717	0.759	0.455	0.078

D3 (truth=445 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.918	0.920	0.295	0.019
scrappie	0.853	0.855	0.392	0.020
pelt	0.903	0.915	0.456	0.036
pelt_cuda	0.902	0.904	0.304	0.024
binseg	0.906	0.908	0.369	0.021
hmm	0.810	0.813	0.545	0.046
bocd	0.911	0.914	0.271	0.019
window	0.909	0.911	0.319	0.024
mad	0.898	0.900	0.433	0.024
gradient	0.664	0.667	0.513	0.039

D4 (truth=485 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.632	0.897	0.237	0.039
scrappie	0.559	0.769	0.247	0.021
pelt	0.619	0.891	0.309	0.041
pelt_cuda	0.617	0.888	0.267	0.035
binseg	0.589	0.839	0.295	0.022
hmm	0.452	0.630	0.307	0.016
bocd	0.629	0.884	0.224	0.020
window	0.598	0.857	0.279	0.035
mad	0.566	0.771	0.373	0.033
gradient	0.083	0.130	0.072	0.000

Per-dataset F1 across 4 metrics — best per column highlighted

D6 (truth=1,167,335 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.811	0.851	0.485	0.077
scrappie	0.014	0.026	0.009	0.003
pelt	0.793	0.833	0.498	0.083
pelt_cuda	0.793	0.833	0.498	0.083
binseg	0.776	0.815	0.496	0.085
hmm	0.357	0.382	0.267	0.055
bocd	0.660	0.692	0.278	0.015
window	0.047	0.058	0.028	0.006
mad	0.004	0.015	0.002	0.001
gradient	0.151	0.166	0.089	0.015

RB_ecoli (truth=12,680 reads)

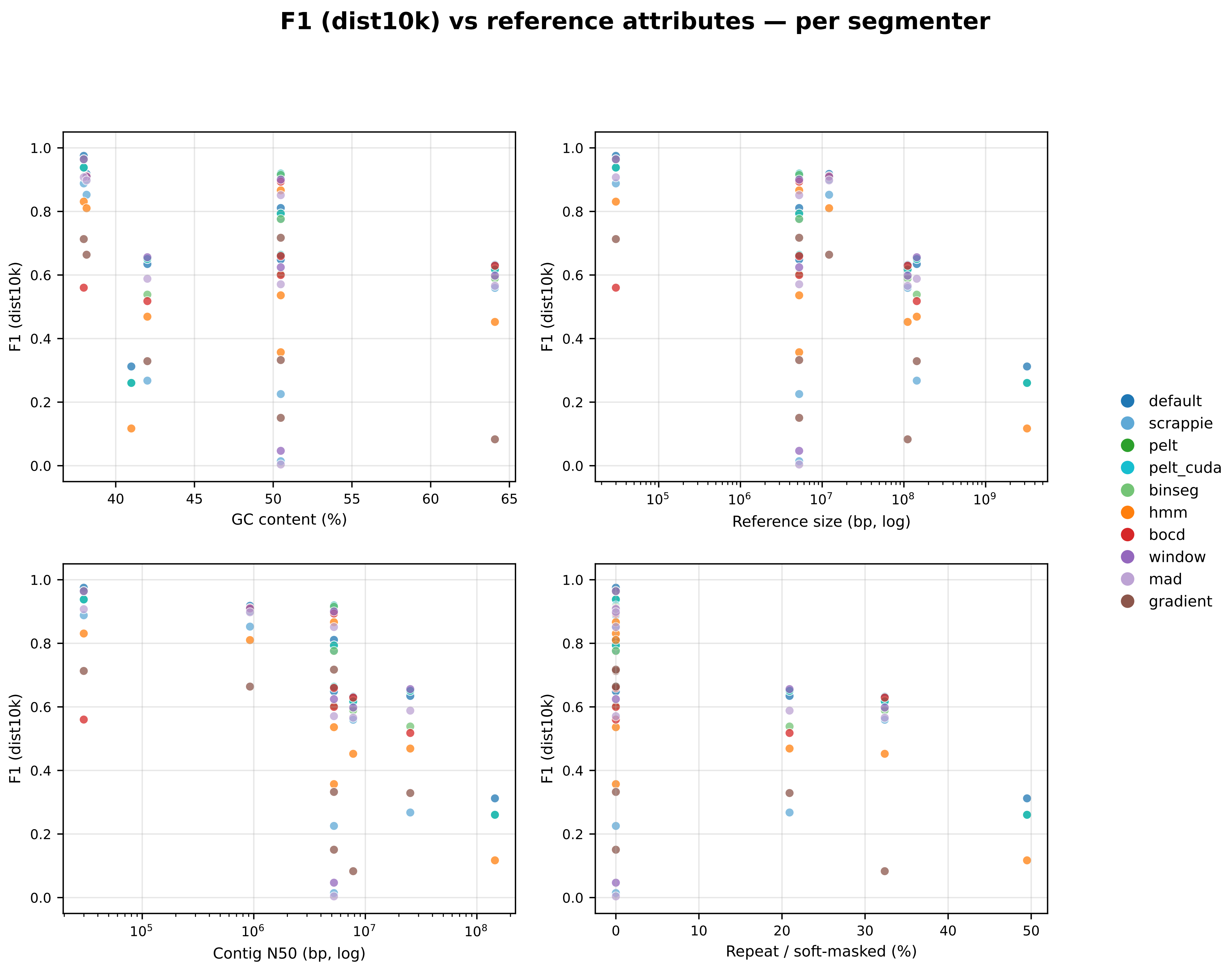
seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.649	0.657	0.419	0.030
scrappie	0.225	0.239	0.130	0.016
pelt	0.663	0.673	0.429	0.031
pelt_cuda	0.663	0.673	0.429	0.029
binseg	0.604	0.612	0.413	0.030
hmm	0.536	0.544	0.456	0.064
bocd	0.600	0.606	0.307	0.012
window	0.624	0.633	0.392	0.029
mad	0.571	0.579	0.414	0.037
gradient	0.332	0.345	0.278	0.039

RB_dmel (truth=11,071 reads)

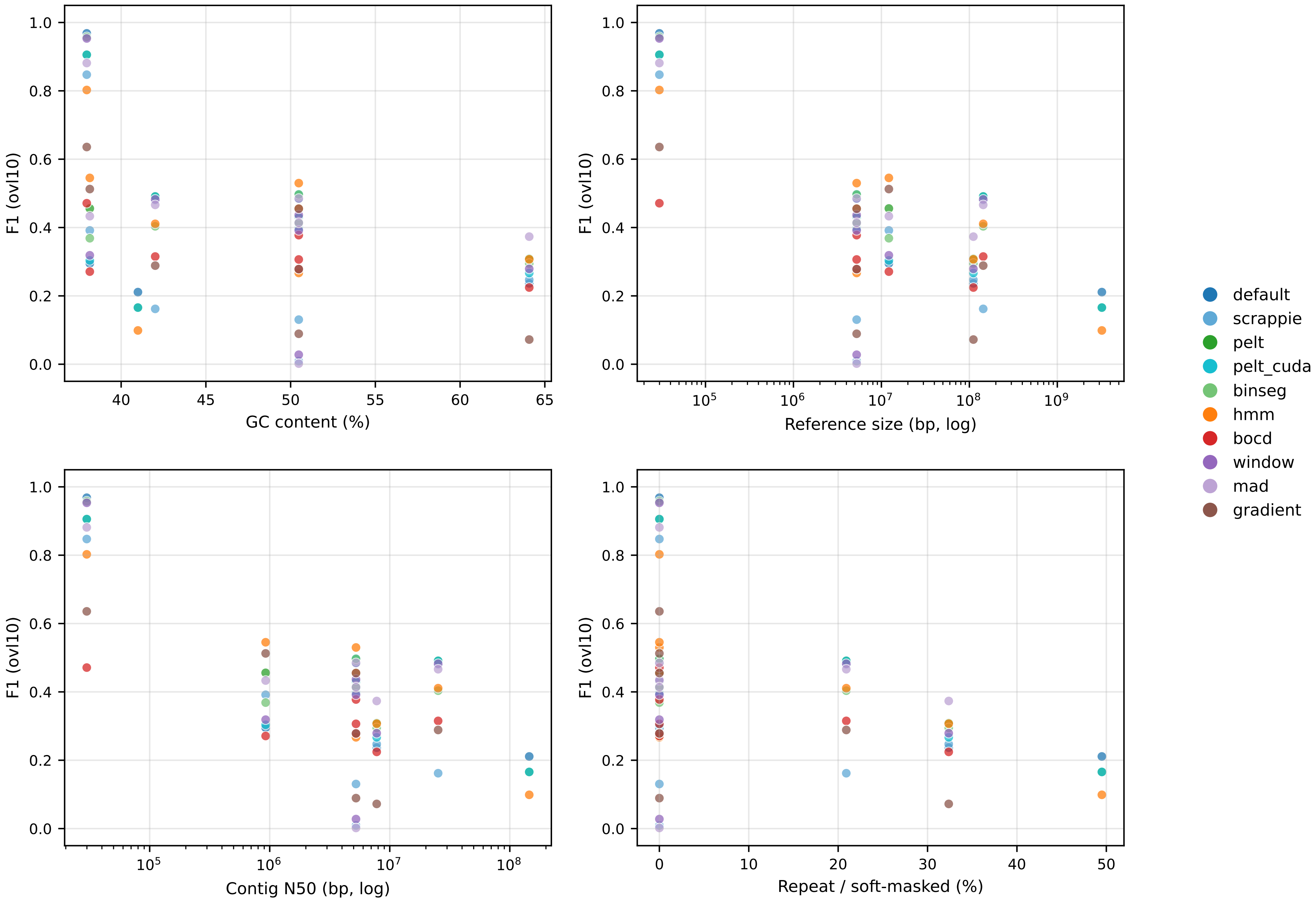
seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.635	0.689	0.480	0.123
scrappie	0.268	0.291	0.162	0.027
pelt	0.651	0.708	0.491	0.127
pelt_cuda	0.650	0.707	0.491	0.127
binseg	0.538	0.586	0.404	0.088
hmm	0.469	0.514	0.411	0.115
bocd	0.518	0.562	0.315	0.034
window	0.656	0.712	0.482	0.132
mad	0.588	0.641	0.466	0.126
gradient	0.329	0.365	0.289	0.074

RB_hsap_full (truth=12,716 reads)

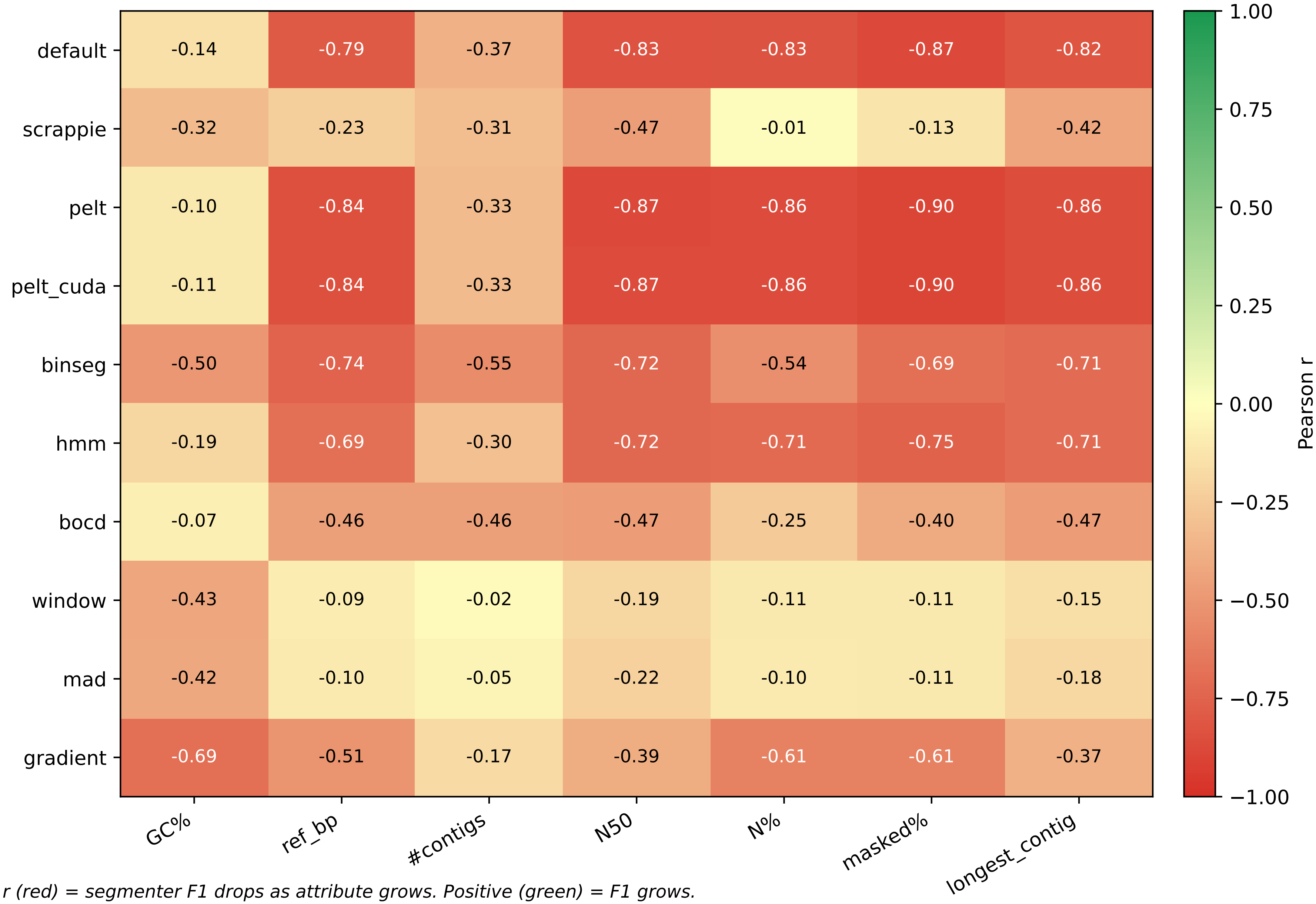
seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.312	0.625	0.211	0.062
pelt	0.260	0.522	0.166	0.045
pelt_cuda	0.260	0.522	0.166	0.045
hmm	0.117	0.232	0.099	0.022


F1 (dist10k) vs reference attributes — per segmenter
F1 (dist10k)
GC content (%)
Reference size (bp, log)
Contig N50 (bp, log)
Repeat / soft-masked (%)
default
scrappie
pelt
pelt_cuda
binseg
hmm
bocd
window
mad
gradient

F1 (ovl10) vs reference attributes — per segmenter

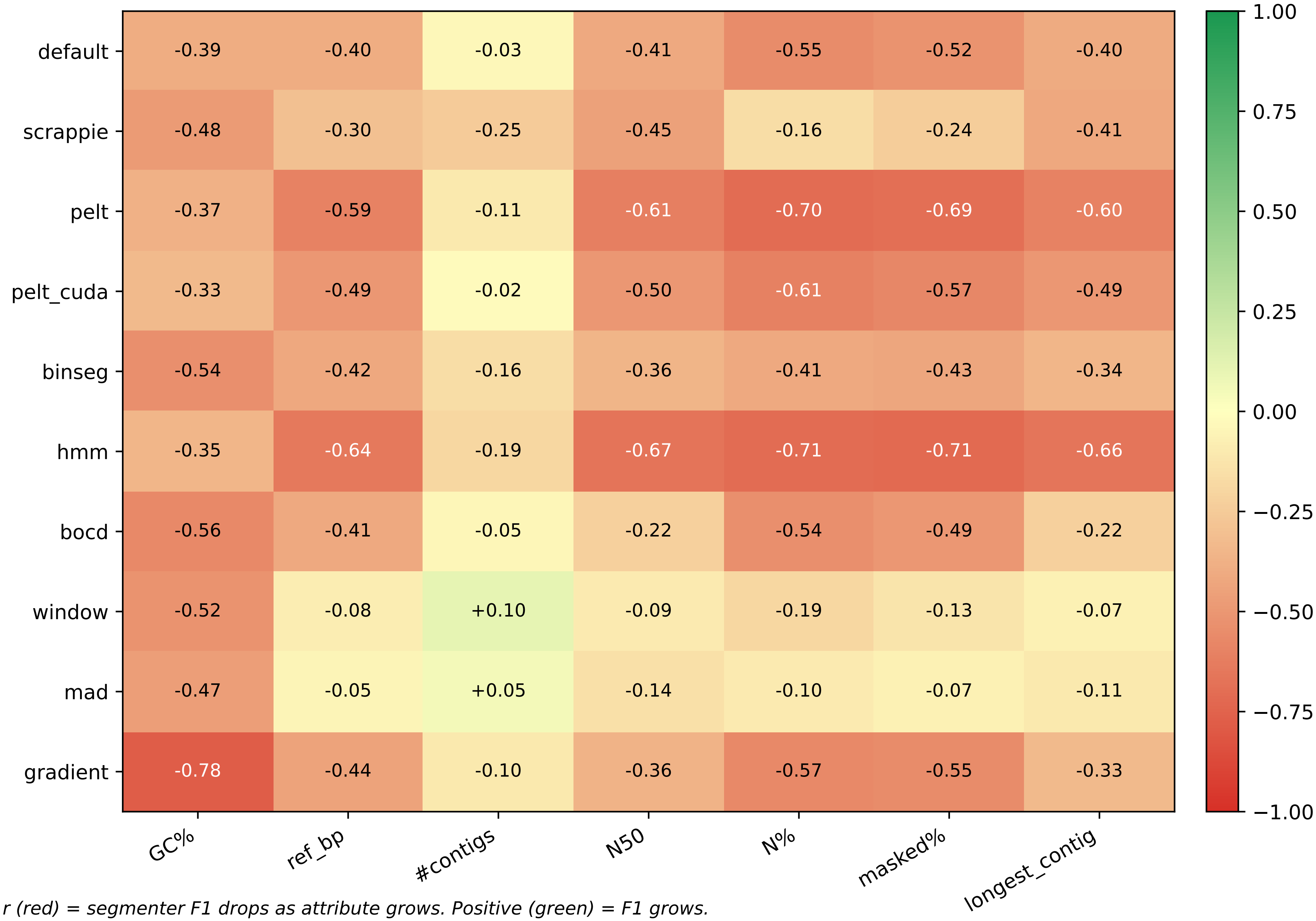


Pearson correlation: F1 (dist10k) vs reference attribute



Negative r (red) = segmenter F1 drops as attribute grows. Positive (green) = F1 grows.
n_datasets = 8. Read with caution (small sample, attributes are correlated).

Pearson correlation: F1 (ovl10) vs reference attribute



Negative r (red) = segmenter F1 drops as attribute grows. Positive (green) = F1 grows.
n_datasets = 8. Read with caution (small sample, attributes are correlated).

CV-best HMM/BOCD config per dataset (3-fold read-level CV, ovl≥0.1 metric)

Dataset	Segmenter	Best config	F1_test	±std	#cfgs
D1	hmm	RH2_HMM_STATES=6 R	0.8675	0.0004	12
D1	bocd	RH2_BOCD_EXP_LEN=6	0.5068	0.0010	6
D2	bocd	RH2_BOCD_EXP_LEN=6	0.3877	0.0004	6
D2	hmm	RH2_HMM_STATES=4 R	0.5381	0.0015	12
D3	hmm	RH2_HMM_STATES=6 R	0.5583	0.0197	12
D4	bocd	RH2_BOCD_EXP_LEN=6	0.0034	0.0002	6
D4	hmm	RH2_HMM_STATES=5 R	0.0058	0.0008	12
D6	hmm	RH2_HMM_STATES=3 R	0.0162	0.0001	1
D6	bocd	RH2_BOCD_EXP_LEN=6	0.3187	0.0009	6
RB_dmel	hmm	RH2_HMM_STATES=6 R	0.4286	0.0064	12
RB_dmel	bocd	RH2_BOCD_EXP_LEN=6	0.3560	0.0105	6
RB_ecoli	bocd	RH2_BOCD_EXP_LEN=6	0.3566	0.0061	6
RB_ecoli	hmm	RH2_HMM_STATES=6 R	0.5040	0.0036	12

Speedup table — default vs tuned-pelt vs pelt_cuda

DS	default wall	pelt wall	pelt_cuda wall	pelt speedup	pelt_cuda speedup	F1 default	F1 pelt	F1 pelt_cuda
D1	1359.1s	10420.1s	1333.5s	0.13×	1.02×	0.975	0.938	0.938
D2	549.1s	—	722.8s	—	0.76×	0.915	0.919	0.919
D3	5.8s	—	4.0s	—	1.47×	0.918	0.903	0.902
D4	13.0s	89.7s	41.9s	0.15×	0.31×	0.632	0.619	0.617
D6	1331.8s	—	9226.0s	—	0.14×	0.811	0.793	0.793
RB_ecoli	54.6s	613.9s	238.6s	0.09×	0.23×	0.649	0.663	0.663
RB_dmel	366.1s	1320.3s	1306.2s	0.28×	0.28×	0.635	0.651	0.650
RB_hsap_full	—	—	—	—	—	0.312	0.260	0.260